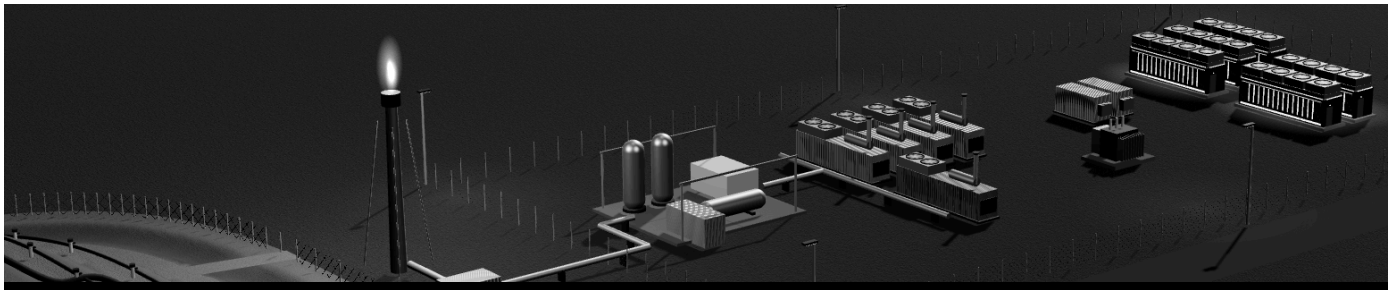
Robin Hood Bay Regional Waste Management Facility, St. John's, Newfoundland and Labrador: a first sizing

What the gas the site now flares could power, how much land it would take, and what it could earn each year.

5.1 MW	5	11 ac	467k
POWER FROM THE GAS	COMPUTE MODULES	LAND NEEDED	MMBTU BOUGHT / YR

GAS SALE, PER YEAR	LAND LEASE, PER YEAR	TOTAL TO THE OWNER, PER YEAR
up to $351k	$11k to $22k	$11k to $373k



YOUR SITE IN OUR LIVE 3D MODEL — ACCRETIONCOMPUTE.COM/SITE-MODEL?SRC=LFG&FLOW=1850&CH4=50

THE MATH

Landfill gas burned in the enclosed flare (estimate, 2024)	1,850 scfm
Methane content (assumed)	50%
Energy in the gas	56.2 MMBtu per hour
Power at a heat rate of 11,000 Btu/kWh	about 5.1 MW
Compute (power less cooling and losses, /1.2)	about 4.3 MW
Compute modules (about 1 MW each)	5
Gas engines (running plus one spare)	5 at 1.6 MW
Land: 5 acres plus 1.2 acres per MW	about 11 acres
Gas bought per year (95% run time)	467k MMBtu
Gas payment, up to $0.75/MMBtu	up to $351k
Land lease at $1,000 to $2,000 per acre per year	$11k to $22k

Landfill gas: about 1,012 Btu per cubic foot of methane. Prices in US dollars; the price band is the one shown on our Site model page.

WHAT IT WOULD LOOK LIKE

- Gas engines on a small parcel near the gas, a battery to smooth the load, and factory-built compute modules leased to an AI company.
- No grid connection and no export: all of the power is used on site.
- We build and run the site. The owner puts in no capital.
- Multi-year gas and land agreements, matched to the tenant's lease, at a fixed price.

WORTH CHECKING

- Measured flow and methane from the City's 2025 feasibility study (Phase 1 report)
- Own-use structure under the Electrical Power Control Act and the Public Utilities Act
- Provincial Certificate of Approval and Environmental Assessment registration
- Federal Landfill Methane Regulations from January 1, 2028: the flare stays as backup

Source: Environment and Climate Change Canada greenhouse gas reporting, 2024 (facility G11386): about 25,600 tonnes of biomass CO2 from burned gas, read as about 1,850 scfm; methane assumed at 50% until the City's Phase 1 report figure is known. This is a first read from public data, not an offer: no cost and no obligation. The real price and terms are set in the gas and land agreements after a site review.